A Biologically-Inspired Framework for Online Mining Spatio-Temporal Dependency Rules from Event Data Streams

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Introduction

Introduction

The paper proposes a biologically inspired framework for continuously mining *Spatio-Temporal Dependency Rules (STDRs)* from event data streams $\mathbf{D}$.

An STDR is a spatially annotated ordered pair of event-type/location tuples:

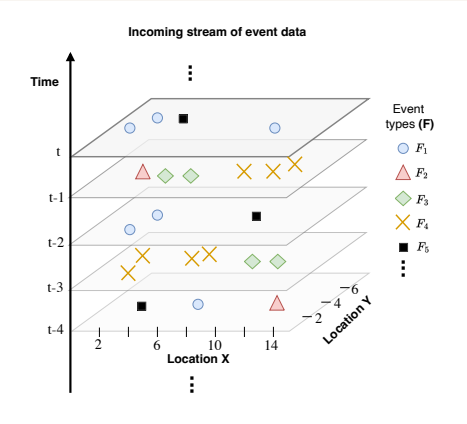
$$\overrightarrow{r} = (F_i, \mathrm{loc}_i) \xrightarrow{w_{ij}} (F_j, \mathrm{loc}_j)$$

- Events of type F_i at loc_i exhibit a **dependency** with subsequent events of type F_j at loc_j.
- The dependency reflects the current state of the data stream $\mathbf{D}$.
- The weight w_{ij} expresses the **significance or strength** of this dependency.

Example STDR

$\overrightarrow{r} = (office,\ location_1) \xrightarrow{w} (subway,\ location_2)$.

Data stream representation



- Let $\mathbf{D} = \{e_1, e_2, \dots\}$ denote a continuous stream of discrete event instances.
- Each event has a type $F \in \mathbf{F}$, a discrete location (e.loc) and a timestamp (e.time).
- Spatial location can be expressed with e.g., two geographical coordinates.

We do not assume any limitations on the number of events in data stream and the number of event types.

Architecture and Learning

Spiking neural network-based architecture (layout)

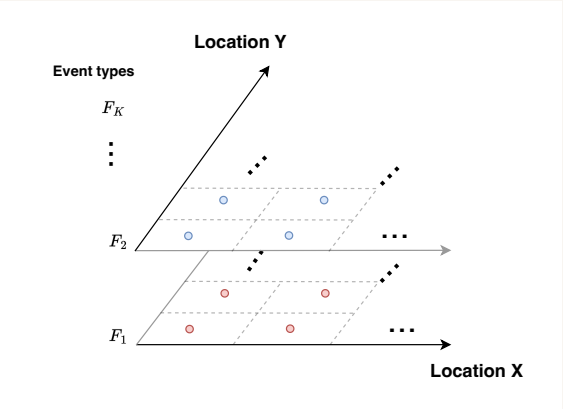


Figure: The proposed architecture of a neural network.

- The data stream $\mathbf{D}$ is used to *train* the proposed neural network
- The set of event types $\mathbf{F}$ *determines* its architecture.
- The spatial region covered by the event instances in $\mathbf{D}$ is partitioned into a grid of $m_1 \cdot m_2$ rectangular cells.
- At the center of each cell, we place a group of neurons, with one neuron corresponding to each event type in $\mathbf{F}$.

Spiking neural network-based architecture (synapses)

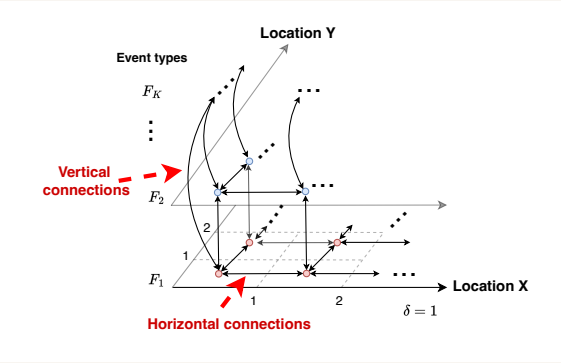


Figure: The proposed architecture of the neural network.

- Within each **spatial** cell, all neurons are fully connected by bidirectional synapses (a pair of directed synapses), one in each direction: for any two neurons n_i and n_j: $n_i \rightarrow n_j$ and $n_j \rightarrow n_i$ (**vertical connections**).
- We also introduce bidirectional synapses between neurons located in different cells if the centers of their corresponding cells are within a user-defined spatial threshold δ (**horizontal connections**).

Event-driven neuronal model

- Each neuron n represents an event type F in a spatial cell c.
- When an event $e \in \mathbf{D}$ occurs, its type $e.F$ and location $e.\mathrm{loc}$ **identify the corresponding neuron**. If $e.F = F$ and $e.\mathrm{loc} \in c$, neuron n emits a spike at the event's timestamp $t = e.\mathrm{time}$.

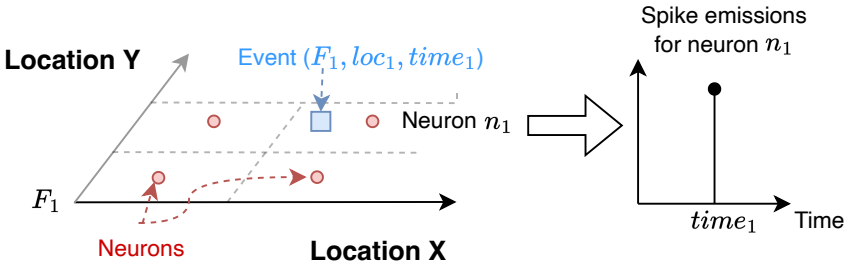


Figure: The proposed neuronal model.

Hebbian-inspired unsupervised learning (rule)

The network's synaptic weights are initialised to zero and updated using an **unsupervised, STDP-inspired Hebbian learning rule**.

$$\frac{dw_{ij}}{dt} = \underbrace{\eta A}_{\text{reinforcement}} - \underbrace{\eta R w_{ij}}_{\text{scaling}} - \underbrace{\lambda w_{ij}}_{\text{forgetting}}$$

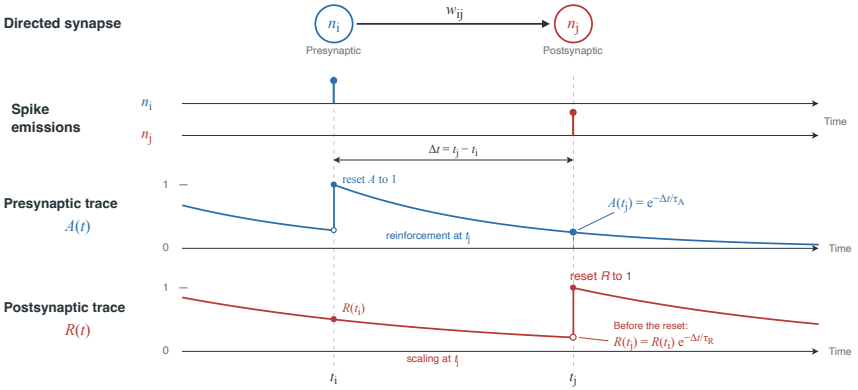
The **presynaptic** A and **postsynaptic** R **traces** decay exponentially between events:

$$A(t + \Delta t) = A(t)e^{-\Delta t/\tau_A}, \qquad R(t + \Delta t) = R(t)e^{-\Delta t/\tau_R}$$

- η controls the learning rate. τ_A and τ_R control the decay timescales of the traces.
- $\lambda > 0$ controls the rate of long-term weight decay.

Hebbian-inspired unsupervised learning (traces)

$$\frac{\mathrm{d}w_{ij}}{\mathrm{d}t} = \underbrace{\eta A}_{\text{reinforcement}} - \underbrace{\eta R w_{ij}}_{\text{scaling}} - \underbrace{\lambda w_{ij}}_{\text{forgetting}}$$



At t_j: update the weight using the current trace values, then reset R to 1.

Upper bound on synaptic weights

Theorem

The upper bound $w_{\max}$ on any synaptic weight w_{ij} is equal to $\frac{\eta}{\lambda}$.

Proof.

(1) Let us rewrite weight change rule as follows: $\frac{dw_{ij}}{dt} = \eta A(t_j) - (\eta R(t_j) + \lambda)\, w_{ij}$,
(2) Next we assume: $A(t) \geq 0$, the maximum of $A(t)$ is $A_{\max} = 1$, $R(t) \geq 0$, $\lambda > 0$, and $\eta > 0$.
(3) Thus, the following inequality is preserved: $\eta R(t) + \lambda \geq \lambda$.
(4) Thus, the following upper bound on $\frac{dw_{ij}}{dt}$ is preserved: $\frac{dw_{ij}}{dt} \leq \eta A_{\max} - \lambda w_{ij}$, which can be further simplified by replacing $A_{\max}$ with 1: $\frac{dw_{ij}}{dt} \leq \eta - \lambda w_{ij}$.
(5) Assume that $w_{\max} = \frac{\eta}{\lambda}$. Then, we have: $\frac{dw_{ij}}{dt} \leq \eta - \lambda w_{ij} \leq \eta - \lambda w_{\max} = 0$.
Therefore, at the boundary $w_{ij} = w_{\max}$, the derivative is non-positive, and the weight cannot increase beyond this value. □

Spatio-temporal dependency rules

Definition (Spatio-Temporal Dependency Rule, STDR)

An STDR is a directed pair of event-type/location tuples $\overrightarrow{r} = (F_i, loc_i) \xrightarrow{w_{ij}} (F_j, loc_j)$.

- (F_i, loc_i) represents the event type and spatial location of exactly **one** neuron n_i.
- w_{ij} is the synaptic weight from n_i to n_j.

Definition (Significance of an STDR)

The *significance of an STDR* $\overrightarrow{r}$, denoted by $\mathrm{sig}(\overrightarrow{r})$, is defined as the weight of the synapse constituting the rule, i.e., $\mathrm{sig}(\overrightarrow{r}) = w_{ij}$.

Definition (Significant STDR)

Let $\theta \in (0, 1)$ be a user-defined significance threshold ratio. An STDR $\overrightarrow{r}$ is called significant if $\mathrm{sig}(\overrightarrow{r}) \geq \theta \cdot w_{\max}$, where $w_{\max} = \dfrac{\eta}{\lambda}$.

Online STDR Mining Algorithm

Online STDR mining with lazy updates

1. Initialise the network with zero weights and traces, and an empty set of significant synapses $\mathbf{W}$.
2. For each incoming event from stream $\mathbf{D}$, "emit a spike" by the neuron corresponding to its event type and spatial cell.
3. Select only the firing neuron's incoming and outgoing synapses and the previously significant synapses for updating (**lazy approach**).
4. Decay the selected weights ($-\lambda w_{ij}$, forgetting) and traces using the time since their last update, update weights according to learning rule.
5. Prune from $\mathbf{W}$ all weights below threshold $\theta \cdot w_{max}$.
6. Report $\mathbf{R}$ being the set of all significant STDRs induced by $\mathbf{W}$.

Computational and memory complexity

- Per-event computational complexity is **excellent**: it scales linearly with the number of currently significant synapses $|\mathbf{W}|$ and $d_{\max}$, the maximum total number of incoming and outgoing synapses of any neuron.
- Memory complexity is more nuanced: it scales linearly with the sum of the total number of synapses in the network $|\mathbf{E}|$ and the number of currently significant synapses $|\mathbf{W}|$.

Time per event

$$O(d_{\max} + |\mathbf{W}|)$$

Total memory

$$O(|\mathbf{E}| + |\mathbf{W}|)$$

$d_{\max}$: maximum total in- and out-synapses for any neuron.
$\mathbf{E}$: all synapses in the network. $\mathbf{W}$: currently significant synapses.

Evaluation

New York City (Foursquare) check-in data stream

227,428

timestamped and geolocated events in stream $\mathbf{D}$.

251

event types $\mathbf{F}$ (social, transportation, and work-related activities).

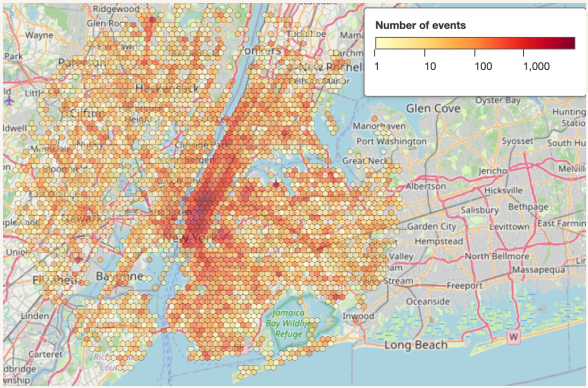
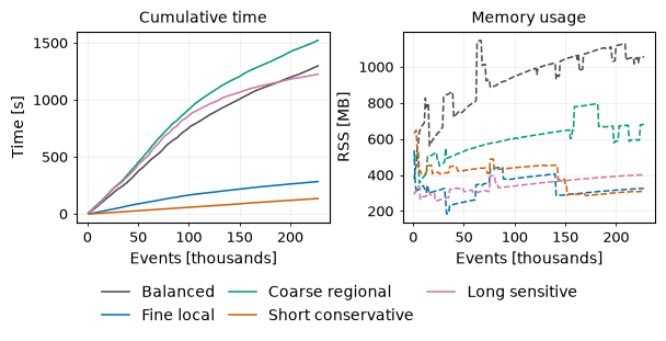


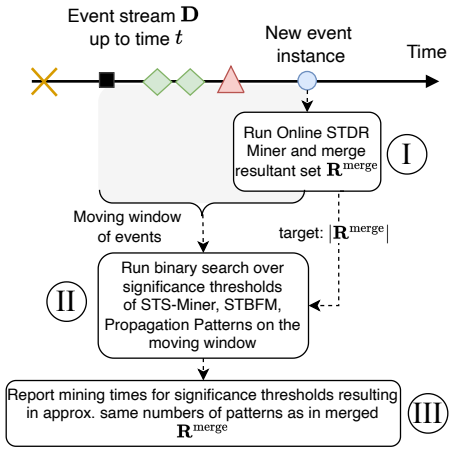
Figure: Density of event locations in the dataset.

Runtime and memory over the full stream



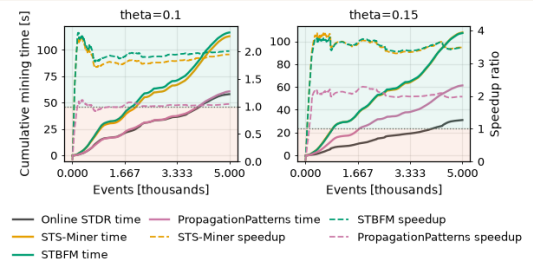
Setup	**Grid**	δ [km]	η	τ_A [min]	τ_R [min]	λ [min^{-1}] ($T_{10\%}$ [min])	θ
Balanced	10×10	5	1	60	12	0.01919 (120)	0.15
Fine local	12×12	4	1	60	12	0.01919 (120)	0.15
Coarse regional	6×6	10	1	60	12	0.01919 (120)	0.15
Short conservative	12×12	4	0.75	30	6	0.03838 (60)	0.25
Long sensitive	12×12	4	1.25	120	24	0.00959 (240)	0.08

Comparison with offline pattern miners



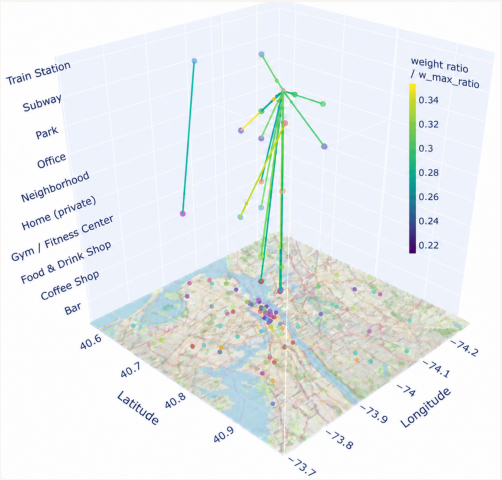
- STS-Miner, STBFM and Propagation Patterns on a 120-minute moving window.
- Patterns restricted to pairs or two-node structures.
- Remove spatial annotations and merge equivalent STDRs.
- Tune baseline thresholds to match the approximate output size.

Mining time against the adapted baselines



Online STDR miner tends to outperform the other approaches. Especially for larger θ values.

Spatial interpretation of the discovered rules



18:00–21:00, $\theta = 0.2$

The snapshot highlights after-work activity dependencies.

The strongest reported rule links an Office cell in Brooklyn with a Subway cell in Manhattan.

The prototype supports event-type selection and filtering of the strongest rules.

Conclusions

Contributions

- An evolving network (framework) that represents spatially explicit dependency rules.
- We defined a new learning rule for the network and provided an upper bound on the possible synapses weights.
- Experiments show a runtime advantage in most tested settings inline with excellent computational complexity.

Current limitations

- STDRs capture only pairwise dependencies.
- Results depend on the spatial grid, neighborhood radius and learning parameters.
- Evaluation covers one application domain.
- The significance of rules can be difficult to interpret.

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Backup

Parameter roles

Parameter	Role
$m_1 \times m_2$	Spatial resolution and number of cell/type neurons
δ	Radius between cell centers that permits connections
η	Learning increment scale
τ_A, τ_R	Duration of the activation and scaling traces
λ	Rate of long-term forgetting
θ	Selection threshold as a fraction of η/λ

Finer grids and broader neighborhoods can increase connectivity and storage costs.

Sources

- P. S. Maciąg (2026). A Biologically-Inspired Framework for Online Mining Spatio-Temporal Dependency Rules from Event Data Streams.
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- Maciąg, Kryszkiewicz and Bembenik (2019). Discovery of Closed Spatio-Temporal Sequential Patterns from Event Data. *KES 2019*.
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- Yang et al. (2014). Modeling User Activity Preference by Leveraging User Spatial Temporal Characteristics in LBSNs. *IEEE TSMC: Systems*.

Code, results and supplementary material: Online_STDR_Miner